Oxford Cambridge and RSA – style practice paper

GCSE (9–1) Computer Science

J277/02 Computational thinking, algorithms and programming

Topic 2.4: Boolean logic

Candidate forename		**Candidate surname**	
Centre number		**Candidate number**	

Practice questions in the style of OCR GCSE Computer Science (J277). Every question is original; the layout, command words, mark allocations and mark scheme follow the real exam format.

Time allowed	About 70 minutes (roughly 1 minute per mark)
Instructions	• Use black ink. HB pencil may be used for graphs and diagrams only. • Answer **all** the questions. • Write your answer to each question in the space provided. • Where the question asks you to write code or pseudocode, you may use any suitable high-level language or the OCR Reference Language; marks are for the logic, not exact syntax. • Do **not** use a calculator or any other electronic device unless told to.
Information	• The total mark for this paper is **64**. • The marks for each question are shown in brackets [ ]. • Quality of extended responses will be assessed in questions marked with an asterisk (*). • Where a question asks you to *state* or *give*, a short answer is enough; *describe* and *explain* need more detail.

Name: _____ Date: _____ Mark: _____ / 64

1 Logic gates are used in computer systems.

(a) Complete the truth table for an **AND** gate. **[2]**

A	**B**	**A AND B**
0	0	
0	1	
1	0	
1	1	

(b) Complete the truth table for an **OR** gate. **[2]**

A	**B**	**A OR B**
0	0	
0	1	
1	0	
1	1	

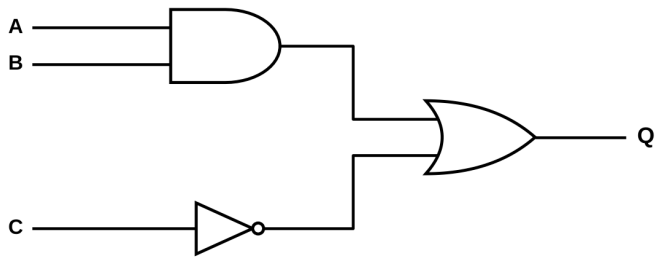
(c) Complete the truth table for a **NOT** gate. **[1]**

A	**NOT A**
0	
1	

(d) State what is meant by a **Boolean** value. **[1]**

(e) Give **one** reason why computers use binary logic gates. **[1]**

2 The diagram shows a logic circuit with inputs **A**, **B** and **C**.



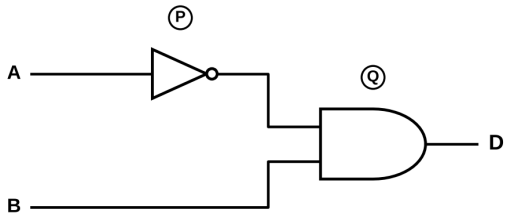
(a) Write the Boolean expression for the output **Q**. **[3]**

(b) Complete the truth table for the circuit. **[4]**

A	B	C	A AND B	NOT C	Q
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

(c) State the output Q when A = 1, B = 0 and C = 1. **[1]**

3 The diagram shows another logic circuit. Two of the gates are labelled **P** and **Q**.



(a) Name the logic gate labelled **P**. **[1]**

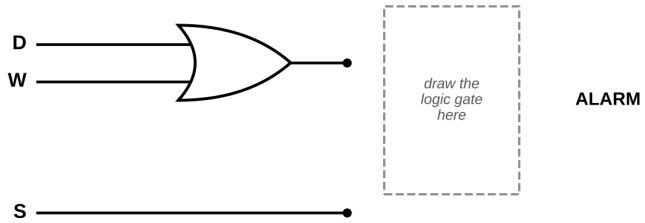
(b) Name the logic gate labelled **Q**. **[1]**

(c) Write the Boolean expression for the output **D**. **[2]**

(d) Complete the truth table. **[3]**

A	**B**	**NOT A**	**D**
0	0		
0	1		
1	0		
1	1		

4 A home alarm system has three inputs. The alarm sounds when the system is switched on (**S** = 1) **and** either the door (**D**) or the window (**W**) is open (= 1).



(a) Write the Boolean expression for the alarm. **[3]**

(b) Complete the logic diagram by drawing the missing logic gate in the dashed box and connecting it to the alarm output. **[2]**

(c) State the output of the alarm when S = 1, D = 0 and W = 1. **[1]**

(d) State the output of the alarm when S = 0, D = 1 and W = 1. **[1]**

5 Boolean logic is used in programming and design.

(a) A security light comes on when it is dark (**Dark** = 1) and either motion is detected (**Motion** = 1) or the manual switch is on (**Switch** = 1). Complete the truth table. **[4]**

Dark	Motion	Switch	Light
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

(b) Draw a logic diagram for the expression **P = NOT (A AND B)**. **[3]**

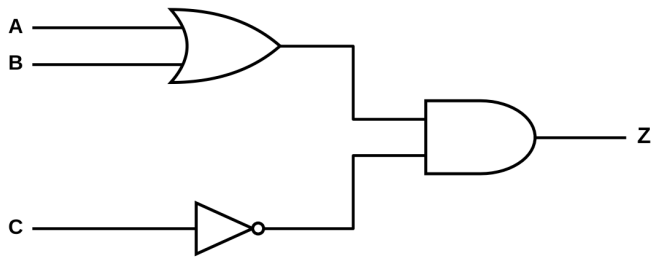
(c) Complete the truth table for **P = NOT (A AND B)**. **[2]**

A	**B**	**A AND B**	**P**
0	0		
0	1		
1	0		
1	1		

(d) A programmer writes the condition: **if age >= 18 AND hasTicket == True then**. State the values of age and hasTicket for which the condition is true. **[2]**

(e) State the result of the expression **NOT (True OR False)**. **[1]**

6 The diagram shows a logic circuit with inputs **A**, **B** and **C**.



(a) Write the Boolean expression for **Z**. **[3]**

(b) Complete the truth table for the circuit. **[4]**

A	B	C	A OR B	NOT C	Z
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

(c) State the combinations of A, B and C that make Z equal to 1. **[2]**

7 Identifying and writing logic.

(a) Name the logic gate that has this truth table. **[1]**

A	B	Output
0	0	0
0	1	1
1	0	1
1	1	1

(b) Name the logic gate that has this truth table. **[1]**

A	B	Output
0	0	0
0	1	0
1	0	0
1	1	1

(c) Complete the truth table for **Q = (NOT A) OR B**. **[3]**

A	B	NOT A	Q
0	0		
0	1		
1	0		
1	1		

(d) Write a Boolean expression for each description. Use AND, OR and NOT and use brackets where needed. **[6]**

(i) The output is 1 only when X is 1 and Y is 0.

(ii) The output is 1 when A is 0 or B is 0.

(iii) A washing machine runs only if the door (D) is closed and either the start button (S) is pressed or the timer (T) is on.

(e) State the output of this code for the inputs **a = 5, b = 12**. **[2]**

```
if a > 3 AND b < 10 then
    print("X")
elseif a > 3 OR b < 10 then
    print("Y")
else
    print("Z")
endif
```

(f) State the output of the same code for **a = 2, b = 12**. **[1]**

END OF QUESTIONS

MARK SCHEME – Topic 2.4: Boolean logic

How to use this mark scheme: mark your own work in a different colour. A bullet (•) in the Answer column is a separate marking point worth one mark unless stated. A slash (/) separates alternative answers that are equally acceptable. Words in brackets are not needed. **Accept** = acceptable; **Ignore** = neither credited nor penalised; **Do not accept** = contradicts or spoils the answer. For programming questions, any valid solution that works is acceptable and syntax errors are not penalised if the intent is clear. The code in the answers uses the OCR Reference Language style.

Question	Answer	Marks	Guidance
1(a)	• 0, 0, 0, 1 (all four correct) ; • (1 mark for three correct) ;	2 AO1 2.4.1	
1(b)	• 0, 1, 1, 1 (all four correct) ; • (1 mark for three correct) ;	2 AO1 2.4.1	
1(c)	1, 0 ;	1 AO1 2.4.1	
1(d)	a value that is either True or False (1 or 0) ;	1 AO1 2.4.1	
1(e)	• transistors / switches have two states (on / off) ; • the CPU / ALU performs calculations and comparisons using logic gates ;	1 AO1 2.4.1	Any one.
2(a)	• (A AND B) ; • (NOT C) ; • OR combining the two: Q = (A AND B) OR (NOT C) ;	3 AO2 2.4.2	
2(b)	• A AND B column: 0,0,0,0,0,0,1,1 ; • NOT C column: 1,0,1,0,1,0,1,0 ; • Q: 1,0,1,0,1,0,1,1 ; • (all eight values of Q correct) ;	4 AO2 2.4.2	Award 1 mark for each correct column; 1 mark for all of Q correct for the follow-through.
2(c)	0 ;	1 AO2 2.4.2	
3(a)	NOT ;	1 AO1 2.4.2	
3(b)	AND ;	1 AO1 2.4.2	
3(c)	• D = NOT A ; • AND B (D = NOT A AND B) ;	2 AO2 2.4.2	
3(d)	• NOT A column: 1,1,0,0 ; • D column: 0,1,0,0 ; • (all four D values correct) ;	3 AO2 2.4.2	
4(a)	• S AND ; • (D OR W) ; • ALARM = S AND (D OR W) (brackets needed) ;	3 AO3 2.4.2	
4(b)	• AND gate drawn ; • inputs from the OR gate output and S, output to ALARM ;	2 AO3 2.4.2	Mark scheme: an AND gate with the OR output and S as its two inputs and ALARM as the output.
4(c)	1 ;	1 AO2 2.4.2	
4(d)	0 ;	1 AO2 2.4.2	

Question	Answer	Marks	Guidance
5(a)	• Light = Dark AND (Motion OR Switch) ; • output column: 0,0,0,0,0,1,1,1 ; • (1 mark per two correct rows, up to 4) ;	4 AO3 2.4.2	Light = 1 only for 101, 110, 111 with Dark = 1.
5(b)	• AND gate with inputs A and B ; • NOT gate connected to the AND output ; • output labelled P ;	3 AO2 2.4.2	
5(c)	• A AND B: 0,0,0,1 ; • P: 1,1,1,0 ;	2 AO2 2.4.2	
5(d)	• age is 18 or more ; • hasTicket is True ;	2 AO2 2.4.2	
5(e)	False ;	1 AO2 2.4.2	
6(a)	• (A OR B) ; • NOT C ; • Z = (A OR B) AND (NOT C) ;	3 AO2 2.4.2	
6(b)	• A OR B: 0,0,1,1,1,1,1,1 ; • NOT C: 1,0,1,0,1,0,1,0 ; • Z: 0,0,1,0,1,0,1,0 ; • (all Z values correct) ;	4 AO2 2.4.2	
6(c)	• any two of 010, 100, 110 ; • (all three for 2 marks) ;	2 AO3 2.4.2	Accept ABC = 010, 100, 110 (A = 0, B = 1, C = 0 etc.).
7(a)	OR ;	1 AO1 2.4.1	
7(b)	AND ;	1 AO1 2.4.1	
7(c)	• NOT A: 1,1,0,0 ; • Q: 1,1,0,1 ; • (all four Q values correct) ;	3 AO2 2.4.2	
7(d)	• (i) X AND (NOT Y) ; • (ii) (NOT A) OR (NOT B) ; • (iii) D AND (S OR T) ; • (2 marks for each correct expression; 1 mark for a partly correct one, e.g. correct gates but wrong brackets) ;	6 AO3 2.4.2	
7(e)	• Y ; • a > 3 is True and b < 10 is False so AND is False, OR is True ;	2 AO2 2.4.2	
7(f)	Z ;	1 AO2 2.4.2	

Where marks are lost in Topic 2.4

Use this page after marking. Tick the ones that cost you marks, then redo those questions closed-book.

- ☐ **Learn the three truth tables** — AND needs both inputs 1; OR needs at least one 1; NOT flips the input.
- ☐ **Work column by column:** add a column for each gate's output before filling in the final output. This avoids mistakes with 3 inputs.
- ☐ **Eight rows for three inputs:** count in binary 000 to 111 so no row is missed.
- ☐ **Brackets matter:** A AND (B OR C) is not the same as (A AND B) OR C. Write them exactly as the circuit connects.
- ☐ **Drawing gates:** AND = D-shape, OR = curved pointed shape, NOT = triangle with a small circle. Always label inputs and output.
- ☐ **Expressions from a description:** 'and' → AND, 'or' → OR, 'not' → NOT. Underline the key words in the question first.
- ☐ **In programs:** use AND / OR / NOT inside if and while conditions the same way as in logic circuits.